Cade Brown

Agentic performance optimization | Compilers & DSLs | GPU numerical libraries
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Summary

Senior engineer working on CUDA math libraries, compiler integration, and agent-driven kernel development, with research in numerical computing and scientific machine learning.

Technical Focus

CUDA, C++, Python, GPU performance engineering, numerical software, compiler and IR development, JIT/LTO, MLIR, LLVM, SIMT programming, NEON, SVE, scientific computing

Experience

Senior CUDA Math Libraries Engineer | NVIDIA
March 2025–Present | Ann Arbor, MI (Remote)

- Developed a framework for AI-agent-driven CUDA kernel generation, profiling, optimization, modification, and autotuning.
- Developed modular, agent-friendly domain-specific languages for algebraic code transformations and code reuse.
- Developed CUDA Tile integration and generalized autotuning with SIMT interoperability, cuTile Python, NVCC, LTO IR, and compiler technologies.
- Built developer productivity tooling including harnesses, CLIs, reusable setups, sandboxing, onboarding guides, and GPU CI monitoring and triage.
- Studied GPU I/O topologies, algorithmic changes, and simulated performance to inform hardware/software co-design.

CUDA Math Library Engineer | NVIDIA
August 2023–March 2025 | Ann Arbor, MI (Remote)

- Optimized CUDA math-library and cuFFT performance through memory-bandwidth work, asynchronous I/O strategies, and code generation.
- Used JIT LTO to generate optimized kernels at runtime for particular user problems while reducing required precompiled kernels.
- Optimized NVPL CPU math libraries on Grace CPUs with NEON and SVE vectorization.

Experience (continued)

Research Assistant (HPC) | Innovative Computing Laboratory, University of Tennessee, Knoxville

August 2019–May 2023 | Knoxville, TN

- Developed GPU compute kernels for MAGMA and SLATE for scientific and machine-learning applications on supercomputers.
- Optimized MAGMA BLAS for AMD GPUs; related work appears in a coauthored IEEE HPEC 2020 paper.
- Designed SABATH, a platform for reproducible scientific machine-learning applications following FAIR principles; related work appears in a coauthored IEEE HPEC 2022 paper.

Software Architect | InterGauge

August 2022–April 2023 | Knoxville, TN

- Provided technical leadership for an education-technology startup focused on interactive educational AI assistants and classroom focus metrics.

Machine Learning Intern | NVIDIA

May 2022–September 2022 | Remote

- Researched hardware acceleration for neural-network training and inference through DSLs, intermediate representations, kernel generation, and polyhedral GPU compilation.
- Trained and evaluated models for code generation and optimization-pass selection.

Additional Experience

Research Assistant (HCI) | University of Tennessee, Knoxville

August 2021–January 2022 | Knoxville, TN

- Developed Avocat, a terminal tool that interprets error output and suggests shell-command solutions using generative text models and semantic vector search.

Machine Learning Consultant | Qardian Labs

January 2020–April 2020 | Knoxville, TN

- Trained and validated a model for heart-disease identification and anomaly detection, with a web interface for risk assessment and further-investigation decisions.

Additional Experience (continued)

Research Intern (HPC) | Oak Ridge National Laboratory
May 2017–August 2018 | Knoxville, TN

- Contributed with Thomas Hill to SimpleSummit, a portable NVIDIA Jetson cluster used to explain distributed computing near Summit at ORNL.
- Developed FractalExplorer, a distributed GPU-accelerated interactive visualization for Mandelbrot and other fractals on SimpleSummit.

Software Consultant | Agilaire LLC

- Designed and implemented pilog, a hardware/software air-quality data logger with sensor collection, time-series storage, visualization, and programmatic access.

Education

Bachelor's degree, Computer Science | University of Tennessee, Knoxville
2019–2023

Publications

Piotr Luszczek and Cade Brown. Surrogate ML/AI Model Benchmarking for FAIR Principles' Conformance. IEEE HPEC, 2022.

Cade Brown, Ahmad Abdelfattah, Stanimire Tomov, and Jack Dongarra. Design, Optimization, and Benchmarking of Dense Linear Algebra Algorithms on AMD GPUs. IEEE HPEC, 2020.

Research

Cade Brown. OEIS A267263, 2016.

SMCEFR satellite-data contribution with Piotr Luszczek: a 1024 × 1024 RGB subset derived from Sentinel-3 OLCI EFR data; Cade Brown is identified as corresponding author.

Selected Projects

mycc: public C-subset compiler and reimplementation using Bison, Flex, and LLVM, with an optional ORC JIT path.

kscript: Dynamic language implemented in C, with arrays, linear algebra, FFTs, and a WASM REPL.

CARVE: a collaborative RISC-V emulator and visualizer for the web, developed with Gregory Croisdale under Stephen Marz.